AWS OBSERVABILITY

Amazon CloudWatch Complete Reference Notes

Unified observability for metrics, logs, alarms, dashboards, and advanced insights across your entire AWS infrastructure and applications.

Last updated: April 3, 2026 | 19 core topics | Reference & Best Practices

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1 Introduction

Modern cloud infrastructure generates millions of data points hourly — CPU utilization fluctuating across hundreds of instances, API calls succeeding or failing thousands of times per second, application logs streaming continuously from distributed microservices, and database queries executing with varying latency. Without comprehensive monitoring, organizations operate blind.

Amazon CloudWatch provides unified observability across AWS infrastructure, applications, and services, collecting metrics, logs, and traces to transform raw data into actionable insights that prevent outages, reduce troubleshooting time from hours to minutes, and enable proactive optimization.

> **THE COST OF INADEQUATE MONITORING**
>
> Average application downtime costs
> **$300,000 PER HOUR**
> for enterprise organizations, with 80% of outages caused by preventable issues that monitoring could detect early.

Traditional monitoring approaches — installing agents on every server, configuring custom dashboards per application, manually correlating logs across systems — cannot scale to cloud environments where infrastructure changes by the minute. CloudWatch eliminates this complexity through:

- Automatic metric collection from **80+ AWS services**
- Centralized log aggregation with powerful query capabilities
- Intelligent alarms with anomaly detection
- Customizable dashboards providing real-time visibility

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2 CloudWatch Architecture

Core Components & Data Sources

```
Data Sources ├── AWS Services (automatic metrics) │ ├── EC2: CPU, Network, Disk │
├── RDS: Connections, IOPS, CPU │ ├── Lambda: Invocations, Duration, Errors │ ├──
DynamoDB: ConsumedCapacity, Throttles │ └── 80+ services — no configuration needed
│ ├── Custom Metrics (application-generated) │ ├── Business metrics (orders/min) │
├── Custom performance metrics │ └── External system metrics │ └── Logs
(centralized aggregation) ├── Application logs ├── AWS service logs (VPC Flow,
CloudTrail) └── Lambda logs (automatic)
```

CloudWatch Services Overview

Service	Description	Key Feature
Metrics	Time-series data points with namespace organization	15-month retention, dimensions, statistics
Logs	Centralized log aggregation from all sources	Log groups, streams, metric filters, Insights
Alarms	Metric-based threshold monitoring	Composite alarms, anomaly detection, actions
Dashboards	Visualization across regions & accounts	Auto-refresh, shareable URLs
Logs Insights	Fast interactive log query engine	No indexing required, rich query language
Container Insights	ECS/EKS/Kubernetes monitoring	Pod, task, and node-level metrics
Lambda Insights	Enhanced Lambda function monitoring	Cold starts, actual memory usage, CPU time
Contributor Insights	Identify top contributors to metrics	Top talkers, error sources, busiest routes
Application Insights	Automated application dashboards	Anomaly detection for .NET, SQL, Java

Data Flow

```
AWS Service → CloudWatch Metrics → Alarms → Actions (SNS, Lambda) → Dashboards (visualization) → API (programmatic access)
Application → CloudWatch Logs → Metric Filters → Metrics → Alarms → Logs Insights (queries) → Export to S3 (archival)
```

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3 CloudWatch Metrics

Metric Types & Namespaces

Standard metrics follow the namespace pattern `AWS/ServiceName`. Custom metrics use any namespace you define.

Key Service Metrics

Service	Namespace	Key Metrics	Resolution	Cost
EC2	`AWS/EC2`	CPUUtilization, NetworkIn/Out, DiskReadOps, DiskWriteOps, StatusCheckFailed	5 min (standard)	**Free**
EC2 Detailed	`AWS/EC2`	Same metrics, higher granularity	1 min	**$2.10/instance/mo**
RDS	`AWS/RDS`	DatabaseConnections, ReadLatency, WriteLatency, FreeStorageSpace, CPUUtilization	1 min	**Free**
Lambda	`AWS/Lambda`	Invocations, Duration, Errors, Throttles, ConcurrentExecutions	1 min	**Free**
DynamoDB	`AWS/DynamoDB`	ConsumedReadCapacityUnits, ConsumedWriteCapacityUnits, ThrottledRequests	1 min	**Free**

Statistics & Aggregation

Statistic	Description	Best Used For	Example
Sum	Total of all values	Counter metrics (requests, errors)	Total requests in 5 min
Average	Mean of all values	Gauge metrics (CPU, memory)	Average CPU over 5 min
Minimum	Lowest value	Performance thresholds	Fastest response time
Maximum	Highest value	Capacity planning, spike detection	Peak CPU utilization
SampleCount	Number of data points	Data volume analysis	Number of requests logged
p50 / p90 / p99	Nth percentile value	Latency analysis, SLA monitoring	99% of requests faster than X
Extended (p0.0–p100)	Fine-grained percentiles	Extreme-tail analysis	p99.9 for 99.9th percentile

TIP — USE PERCENTILES FOR LATENCY

Use **P99** instead of Average for latency metrics. Average hides slow outliers that negatively impact user experience. *Example: "99% of Lambda functions complete within 2 seconds."*

Aggregation Periods

Period	Use Case	Notes
1 second	High-resolution monitoring	Paid; rapid auto-scaling response
1 minute	Detailed monitoring	Paid for most services
5 minutes	Standard monitoring	Free for most AWS services
1 hour	Long-term trends	Cost optimization analysis
1 day	Capacity planning	Monthly reporting

4 CloudWatch Logs

Log Hierarchy

```
Log Group: /aws/lambda/my-function │ ├── Log Stream: 2025/01/15/[$LATEST]abc123 │
├── Log Event 1: START RequestId: xyz │ ├── Log Event 2: Processing order 123 │ └──
Log Event 3: END RequestId: xyz │ ├── Log Stream: 2025/01/15/[$LATEST]def456 └──
Log Stream: 2025/01/16/[$LATEST]ghi789
```

Level	Description	Key Properties
Log Group	Container for all streams from one application/service	Retention policy, subscription filters, metric filters Convention: `/aws/service/resource-name`
Log Stream	Sequence of events from a single source	Ordered by timestamp; automatically created by source
Log Event	Timestamp + Message	Max 256 KB per event; ingestion up to 5 GB/sec per log group

Retention Policies

By default, CloudWatch logs **never expire**. Always set a retention policy to control costs.

Environment	Recommended Retention	Rationale
Production	30 days	Incident investigation window; export to S3 for longer
Development	7 days	Short-lived debugging sessions
Debug / Verbose	3 days	High-volume, low-value log entries
Compliance	1–10 years (via S3)	Regulatory; use S3 Glacier for cost-efficiency

COMMON LOG GROUPS

Log Group Pattern	Source
`/aws/lambda/<function-name>`	Lambda function logs
`/aws/rds/instance/<name>/error`	RDS error logs
`/aws/ecs/<cluster-name>`	ECS container logs
`/aws/codebuild/<project-name>`	CodeBuild build logs
`/aws/apigateway/<api-id>/<stage>`	API Gateway access logs

Log Ingestion Methods

#	Method	When to Use	Notes
1	**Automatic (AWS Services)**	Lambda, ECS/Fargate, RDS, API Gateway	No agent required; zero configuration for Lambda
2	**CloudWatch Logs Agent** (Legacy)	Existing EC2 / on-premises setups	Being replaced by the unified agent
3	**CloudWatch Unified Agent** (Recommended)	EC2 / on-premises needing metrics + logs	Supports StatsD, collectd; SSM integration
4	**Direct API** (`PutLogEvents`)	Custom applications with full control	SDK-based; supports batch upload
5	**Kinesis Integration**	High-volume real-time streaming scenarios	Kinesis Data Firehose → CloudWatch Logs

Structured Logging Best Practice

Always log in JSON format to enable powerful Logs Insights queries and automated analysis.

```
{
    "timestamp": "2025-01-15T10:30:00.123Z",
    "level": "ERROR",
    "requestId": "abc-123-def-456",
    "userId": "user-789",
    "message": "Database connection failed",
    "error": {
        "type": "ConnectionTimeout",
        "code": "ETIMEDOUT",
        "details": "Connection to db.example.com:5432 timed out"
    },
    "duration": 30000,
    "retry": 3
}
```

LOG FORMAT RULES

- ✓ Use structured logging (JSON)
- ✓ Include timestamp, severity, request ID
- ✓ Add context (user ID, session ID, trace ID)
- ✓ Sanitize sensitive data (PII, credentials)

Metric Filters

Extract CloudWatch metrics directly from log events without publishing separate custom metrics.

```
Log Events → Pattern Matching → Extract Values → Publish Metric
```

- ✓ Test patterns with sample logs first
- ✓ Use default value 0 for count metrics
- ✓ Create separate filters for different patterns
- ✓ Monitor filter match rate (low rate = pattern issue)
- ✗ Don't create too many filters — performance impact

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5 CloudWatch Alarms

Alarm Configuration

Component	Description	Example
Metric	What to monitor	`CPUUtilization`
Statistic	How to aggregate	`Average`
Period	Time window	5 minutes
Threshold	Value triggering alarm	80%
Comparison	Operator	`GreaterThanThreshold`
Evaluation Periods	How many periods to evaluate	2
Datapoints to Alarm	M out of N periods	2 out of 2

EXAMPLE ALARM EVALUATION (CPU > 80%, 2-OF-2 PERIODS)

Period	CPU	Datapoints Breached	State
1	85%	1/2	OK
2	82%	2/2	**ALARM**
3	75%	1/2	**ALARM (still)**
4	70%	0/2	**OK**

Alarm States & Actions

State	Meaning	Available Actions
ALARM	Metric breached threshold	Send SNS notification, execute Auto Scaling policy, stop/terminate/reboot EC2, invoke Lambda via SNS
OK	Metric within threshold	Send recovery notification, scale down Auto Scaling, custom recovery actions
INSUFFICIENT_DATA	Not enough data points	Usually no action; optionally notify for investigation

MISSING DATA TREATMENT

Setting	Missing Data Treated As	When to Use
`notBreaching`	OK (default)	Lenient; normal for intermittent metrics
`breaching`	ALARM	Strict; use for crash detection (app stops logging)
`ignore`	Maintain current state	Temporary gaps expected
`missing`	INSUFFICIENT_DATA	Alert when data disappears

IMPORTANT

If your application crashes, it stops logging. With the default `notBreaching` setting, the alarm will

NOT

trigger. Set missing data to `breaching` for crash detection.

Composite Alarms

Composite alarms combine multiple alarms with AND/OR/NOT logic to reduce false positives.

EXAMPLE 1 — APPLICATION HEALTH CHECK

```
cloudwatch.put_composite_alarm(
    AlarmName='ApplicationUnhealthy',
    AlarmRule='ALARM(HighErrorRate) AND (ALARM(HighLatency) OR ALARM(LowSuccessRate))'
    ActionsEnabled=True,
    AlarmActions=['arn:aws:sns:region:account:critical-alerts']
)
```

Sub-Alarm	Condition
HighErrorRate	Error rate > 5%
HighLatency	p99 > 2000 ms
LowSuccessRate	Success rate < 95%

Composite rule: `ALARM(HighErrorRate) AND (ALARM(HighLatency) OR ALARM(LowSuccessRate))`

EXAMPLE 2 — DATABASE DEGRADATION

Sub-Alarm	Condition

HighCPU	CPU > 80%
HighConnections	Connections > 90% max
LowFreeMemory	Memory < 20%

Composite rule: `ALARM(HighCPU) AND ALARM(HighConnections) AND ALARM(LowFreeMemory)`

COMPOSITE ALARM OPERATORS

- **AND** — All conditions must be true
- **OR** — Any condition is true
- **NOT** — Invert a condition
- **Parentheses** — Group conditions for precedence

> **IMPACT**
>
> Composite alarms reduce false positive rate by 90%+, significantly reducing alert fatigue while maintaining visibility into actual issues.

Alarm Math

Use metric math expressions to alarm on derived values — no need to publish additional custom metrics.

```
# Error Rate Alarm: trigger when Lambda error rate exceeds 5%
Expression: errors / invocations * 100 > 5

Where:
  errors      = Lambda Errors metric
  invocations = Lambda Invocations metric
```

---

6 CloudWatch Logs Insights

Fast, interactive log analysis engine with a purpose-built query language. No indexing configuration required.

Query Syntax

Command	Purpose
`fields`	Select fields to display
`filter`	Filter log events by condition
`stats`	Aggregate data with functions
`sort`	Order results
`limit`	Restrict result count
`parse`	Extract values from log strings

Common Query Examples

1. COUNT ERRORS PER HOUR

```
fields @timestamp, @message
| filter @message like /ERROR/
| stats count() as error_count by bin(1h)
```

2. TOP 10 ERROR MESSAGES

```
fields @message
| filter level = "ERROR"
| stats count() as occurrences by @message
| sort occurrences desc
| limit 10
```

3. AVERAGE LATENCY PER ENDPOINT

```
fields @timestamp, endpoint, duration
| stats avg(duration) as avg_latency by endpoint
| sort avg_latency desc
```

4. PARSE JSON LOGS — FILTER 5XX ERRORS

```
fields @timestamp, @message
| parse @message '{"level":"*","status":*,"duration":*}' as level, status, duration
| filter status >= 500
| stats avg(duration) as avg_error_duration
```

5. FIND SLOW REQUESTS (> 3 SECONDS)

```
fields @timestamp, requestId, duration
| filter duration > 3000
| sort duration desc
| limit 100
```

6. ERROR RATE CALCULATION

```
| stats count() as total,
       sum(status >= 500) as errors
| fields errors / total * 100 as error_rate
```

Aggregation Functions

Category	Function	Description
Aggregation	`count()`	Count records
	`sum(field)`	Sum values
	`avg(field)`	Average
	`min(field)`	Minimum
	`max(field)`	Maximum
	`stddev(field)`	Standard deviation
String	`strlen(field)`	String length
	`concat(f1, f2)`	Concatenate fields
	`trim(field)`	Remove whitespace
	`lower(field)`	Lowercase
	`upper(field)`	Uppercase
Date/Time	`bin(interval)`	Time buckets: 1m, 5m, 1h, 1d
	`earliest(@timestamp)`	Earliest time in group
	`latest(@timestamp)`	Latest time in group

QUERY COST

Logs Insights charges **$0.005 PER GB SCANNED**. Optimize by: narrowing time ranges, applying `filter` early, selecting only needed `fields`.

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7 CloudWatch Dashboards

Dashboard Types

Type	Description	Best For
Standard	Custom multi-widget, multi-region/account dashboards	Production environments, SOC views
Automatic	Service-specific dashboards with pre-configured best-practice widgets	Quick service health checks

Widget Types

#	Widget	Description	Best Used For
1	**Line Graph**	Time-series with multiple metrics and annotations	Trend analysis, capacity planning
2	**Number Widget**	Single metric, large readable display with color-coded thresholds	KPI summaries, real-time status
3	**Gauge Widget**	Progress toward threshold; red/yellow/green indicators	Utilization percentage
4	**Bar Chart**	Compare multiple metrics side-by-side	Cross-service or cross-region comparison
5	**Pie Chart**	Percentage distribution (2–5 metrics max)	Resource proportions
6	**Log Widget**	Recent log events or Logs Insights query results; live tail	Real-time log monitoring

7	**Alarm Widget**	Status of multiple alarms at a glance	Operations overview, NOC display
8	**Text Widget**	Markdown-formatted documentation, instructions, and links	Runbook links, context, annotations

Example Production Dashboard Layout

Row	Metrics / Widgets	Widget Type
Row 1 — Business	Orders per minute, Revenue per hour, Active users	Number, Number, Gauge
Row 2 — App Health	Error rate (red threshold), Latency p99, Success rate	Line, Line, Gauge
Row 3 — Infrastructure	EC2 CPU, RDS connections, Lambda throttles	Line, Line, Number
Row 4 — Ops	Critical alarms, Recent deployments with links	Alarm, Text

Dashboard Best Practices

- ✓ Place most critical metrics at the top
- ✓ Group related metrics together
- ✓ Use consistent time ranges across all widgets
- ✓ Include key SLA metrics and alarm status
- ✓ Use color coding (red/yellow/green thresholds)
- ✓ Add Text widgets with links to runbooks
- ✓ Generate shareable links for external stakeholders (expiry: 3h–30d)

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8 Advanced Insights Services

Container Insights (ECS / EKS / Kubernetes)

Dimension	Metrics Collected
Container / Pod	CPU utilization, memory utilization, network bytes in/out
Task / Service	Disk I/O, container restarts
Node / Cluster	Aggregated resource utilization, capacity trends

Available dimensions: `ClusterName`, `ServiceName`, `TaskDefinitionFamily`, `PodName`, `Namespace`. Provides automatic dashboard creation with pod, container, and node-level views.

Lambda Insights

Provides metrics beyond what standard Lambda monitoring offers:

Metric	Description	Use Case
`cpu_total_time`	CPU time consumed by function	Detect CPU-bound functions
`memory_utilization`	Actual memory used vs. allocated	Right-sizing, cost optimization
`init_duration`	Cold start initialization time	Cold start optimization
`tmp_used`	/tmp directory usage	Storage limit monitoring

COST OPTIMIZATION EXAMPLE

Function allocated 1024 MB → actual usage 256 MB → reduce to 512 MB → **50% COST REDUCTION** on invocations.

Contributor Insights

Identify top contributors to metrics and logs — who or what is consuming the most resources.

Use Case	Example
Top talkers	IP addresses with highest request volume
Top errors	Which endpoints are failing most frequently
Heaviest users	Users consuming most API capacity

Busiest routes	Traffic patterns by URL path

Built-in rules cover DynamoDB top partition keys, VPC Flow Logs top talkers, and Route 53 query volume by domain.

9 Cost Optimization

CloudWatch Pricing Reference

CUSTOM METRICS
$0.30
per metric / month
First 10 free

LOG INGESTION
$0.50
per GB ingested

LOG STORAGE
$0.03
per GB-month

LOGS INSIGHTS
$0.005
per GB scanned

DASHBOARDS
$3.00
per dashboard/mo
First 3 free

STANDARD ALARMS
$0.10
per alarm / month

HI-RES ALARMS
$0.30
per alarm / month

COMPOSITE ALARMS
$0.50
per alarm / month

Optimization Strategies

#	Strategy	Action	Savings Potential
1	**Log Retention Policies**	Set 30d prod / 7d dev / 3d debug (vs. default indefinite)	Up to 90%
2	**Export to S3**	Move logs older than 30 days to S3 Glacier ($0.004/GB vs $0.03/GB)	87%

3	**Use Metric Filters**	Extract metrics from logs instead of publishing custom metrics	$0.30/metric/mo saved
4	**Consolidate Metrics**	Reduce to 5 key metrics per instance; enable detailed on-demand	80%+ metric spend
5	**Anomaly Detection**	Fewer false alarms = fewer SNS actions + less investigation time	Indirect cost savings
6	**Delete Unused Resources**	Orphaned log groups, unused dashboards, inactive alarms	Variable

Before vs. After Optimization Example

Resource	Before	After	Savings
Custom Metrics	1,000 → $300/mo	200 → $60/mo	$240
Log Storage	Indefinite → growing	30-day retention → $3/mo	—
Dashboards	20 → $51/mo	5 → Free	$51
Alarms	500 → $50/mo	100 consolidated → $10/mo	$40
Total	**$404/mo**	**$73/mo**	**$331/mo (82%)**

82% **Monthly cost reduction** achievable through proper log retention policies, metric consolidation, dashboard pruning, and alarm consolidation — from $404/mo to $73/mo.

10 Tips & Best Practices

Metrics Best Practices

TIP 1

High-Resolution Metrics for Real-Time Monitoring. 1-second resolution enables rapid auto-scaling response — essential for spiky workloads.

TIP 2

Publish with Consistent UTC Timestamps. Avoids daylight saving issues and timezone confusion across global teams.

TIP 3

Use Dimensions Wisely. Add dimensions for filtering (Environment, Region) — but avoid too many; metric combinations multiply cost.

TIP 4

Leverage Metric Math for Derived Metrics. Calculate error rates and utilization percentages in alarms — no extra custom metrics needed.

TIP 5

Use Percentiles (p99) for Latency. Average hides outliers that degrade user experience. p99 latency is more meaningful for SLA monitoring.

Logging Best Practices

TIP 6

Implement Structured Logging (JSON). Enables powerful Logs Insights queries and automated analysis.

TIP 7

Include Request / Trace IDs in All Logs. Critical for correlating events across distributed microservices during troubleshooting.

TIP 8

Log at Appropriate Levels. DEBUG for development, INFO for production, ERROR always — avoids log volume explosion.

TIP 9

Use Log Sampling for High-Volume Systems. Log 1–10% of successful requests, 100% of errors — reduces costs while maintaining visibility.

TIP 10

Sanitize Sensitive Data Before Logging. Never log PII, credentials, or credit card numbers — compliance violations and security risks.

Alarm Best Practices

TIP 11

Use Composite Alarms to Reduce False Positives. Require multiple conditions — 90%+ reduction in false positives.

TIP 12

Set Appropriate Evaluation Periods. 2–3 periods prevents transient spikes from triggering alarms — balances responsiveness and stability.

TIP 13

Configure Actions for Each Alarm State. ALARM: page on-call. OK: send recovery notification. Complete lifecycle management.

TIP 14

Use Anomaly Detection for Patterned Metrics. ML learns daily/weekly patterns automatically — no manual threshold tuning needed.

TIP 15

Test Alarms Before Production. Manually set alarm state to verify actions — prevents "alarm doesn't work" surprises during real incidents.

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11 Common Pitfalls & Remedies

Pitfall 1 — Excessive Log Retention Costs

Problem: CloudWatch log storage costs grow unexpectedly, consuming a significant portion of the AWS bill.

Why it happens:

- Default retention is **Never expire** (indefinite)
- High-volume applications logging extensively
- Retention never reviewed after initial setup
- Debug logs retained unnecessarily
- Forgotten log groups from deleted applications

Real cost example (100 GB logs/month, indefinite retention):

Month	Accumulated Storage	Monthly Cost
Month 1	100 GB	$3
Month 12	1,200 GB	$36
Month 24	2,400 GB	$72 (and growing)

Remedy

1. Audit all log groups — identify retention status
2. Implement retention policies immediately (30/7/3 days)
3. Export old logs to S3 Glacier for long-term compliance
4. Automate lifecycle management via CloudFormation or Lambda
5. Delete log groups from decommissioned applications
6. Use CloudWatch cost allocation tags to track spend by application

```
# Set log group retention policy (Python SDK)
logs.put_retention_policy(
    logGroupName='/application/production',
    retentionInDays=30
)
```

Pitfall 2 — Alarm Threshold Tuning Challenges

Problem: Alarms trigger too frequently (false positives) or not at all (false negatives), making monitoring useless.

Why it happens:

- Static thresholds don't account for workload patterns
- No baseline metric analysis before setting thresholds
- Daily/weekly traffic cycles not considered
- Thresholds based on guesswork, not data

Real example (CPU > 80% alarm):

- Business hours 9am–5pm: 70–90% CPU (normal) → constant false alarms
- Off-hours 6pm–8am: 10–20% CPU → 80% threshold never breaches
- Real issue at 3am: 78% CPU (below threshold) → not detected

Remedy

1. Analyze 30 days of historical data before setting any threshold
2. Use **Anomaly Detection** for metrics with daily/weekly patterns
3. Implement time-based (dayparting) thresholds where needed
4. Test all alarms before enabling actions in production
5. Review and tune alarms quarterly
6. Document threshold rationale for future reference
7. Use metric math for derived thresholds (error rate, not raw error count)

12 Summary & Key Takeaways

Amazon CloudWatch provides comprehensive observability across AWS infrastructure and applications through unified metrics, logs, alarms, and dashboards. It automatically collects metrics from 80+ AWS services without configuration, aggregates logs with powerful Logs Insights querying, triggers intelligent alarms with anomaly detection, and visualizes system health through customizable dashboards.

Advanced features include Container Insights for ECS/EKS, Lambda Insights for function optimization, Contributor Insights for top resource consumers, and seamless X-Ray integration for distributed tracing.

Key Takeaways

- **Leverage Automatic Metrics:** 80+ AWS services publish metrics automatically — no agents required for EC2, RDS, Lambda, DynamoDB, and more.
- **Implement Structured Logging:** JSON format enables powerful Logs Insights queries, extracting metrics, calculating aggregations, and analyzing patterns.
- **Use Composite Alarms:** Combine multiple conditions to reduce false positives by 90%; prevents alert fatigue while maintaining real visibility.
- **Enable Anomaly Detection:** ML automatically learns normal patterns and adjusts for daily/weekly cycles — eliminates manual threshold tuning.
- **Optimize Log Retention:** Set 30 days for production, 7 days for development; export to S3 for long-term storage — 75% cost reduction achievable.
- **Monitor SLAs Continuously:** Track availability, latency (p99), and error rate; automate monthly SLA reports with CloudWatch metrics.
- **Use Metric Filters Strategically:** Extract metrics from logs without publishing custom metrics — reduces costs while maintaining visibility.

INTEGRATION SCOPE

CloudWatch integrates throughout AWS — monitoring Lambda functions, tracking API Gateway requests, analyzing WAF logs, aggregating GuardDuty findings, and providing operational dashboards for entire multi-account, multi-region environments.